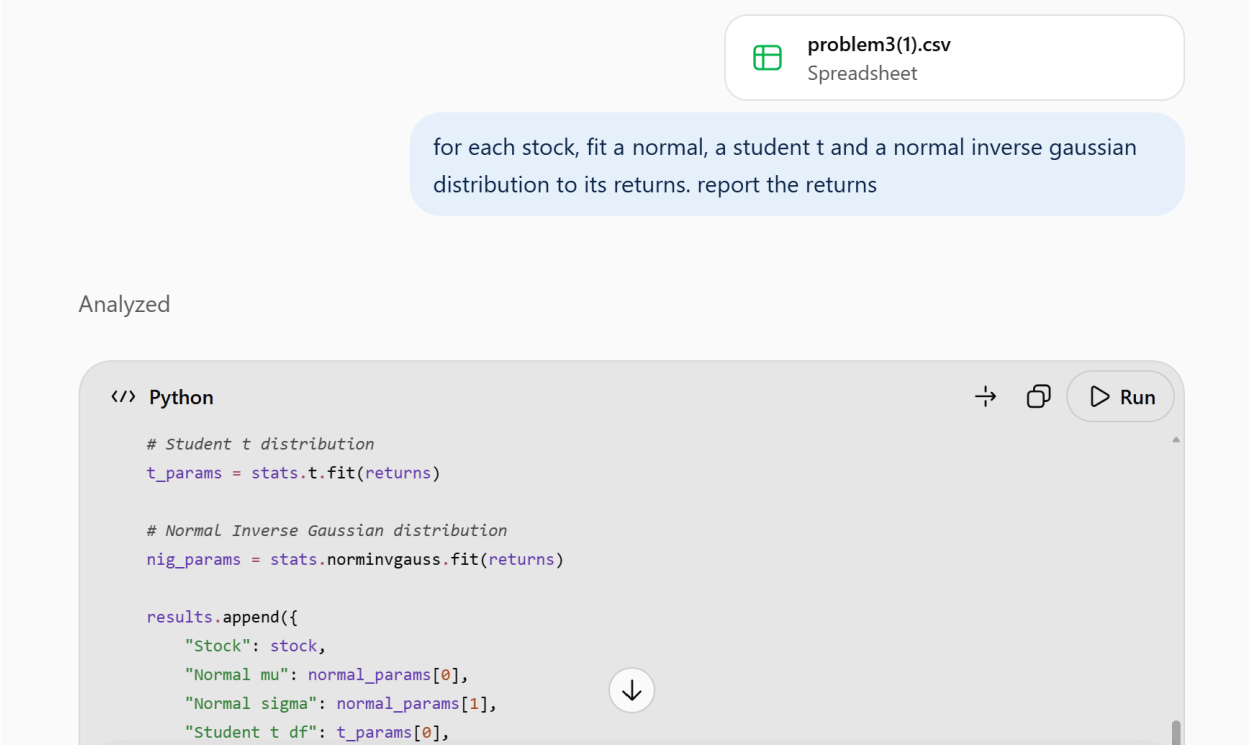
I'm doing a offline midterm exam, I am allowed to use AI to write code, but i cannot use AI to explain the answers or numbers. Start now

- Ready. Send me the exam question, dataset, or coding prompt. I'll help write the code only, keeping explanations and interpretation of answers or numerical results out of the response.

Combine the repaired correlation matrix with the original variances to get the covariance matrix

PV * rl.var_normal(sd_x, mu_x, a), calculate VaR and ES result in percent



(b) Using the distributions you chose in part (a), fit a Gaussian copula and a Student's t copula. Which one fits better? Use my risklib

- Here's the same thing using your risklib. It needs v0.6 or later for `choose_copula`. The NIG margins go through scipy directly, so the NIG problem you hit on your machine won't come up. The values in `choice` are only placeholders: set them to your answer from part (a). Which copula fits better and why is yours to write.

```
last = prices.iloc[-1].drop("Date").astype(float)
hold["price"] = hold["Stock"].map(last)
hold["value"] = hold["Holding"] * hold["price"]
hold["weight"] = hold["value"] / hold["value"].sum()
books = hold.groupby("Book")["Stock"].apply(list).to_dict()
dollars = hold["value"].values
margins, _ = rl.fit_margins(ret)

U = rl.margin_uniforms(ret, margins)
R = rl.copula_corr(U, "spearman")
print("\n(d) min eig of R:", round(rl.min_eigenvalue(R), 4))
Rdf = pd.DataFrame(R, ret.columns, ret.columns)
A, B = books["A"], books["B"]

_, sim = rl.simulate_returns(rl.fit_gaussian_copula(U, R), margins,
100_00, seed=545)
pnl = pd.DataFrame(sim * dollars, columns=ret.columns)
cop = rl.group_var_es(pnl, books, 0.05)
print("\n(e) Gaussian copula\n", cop);
```

verify this code for question (3):Using the copula you chose in part (b), simulate the portfolio with at least 5,000 draws.Calculate the absolute VaR and ES at the 95% level, in dollars, for each stock, each book,and the total portfolio. Present the results in a table.

Your code has five problems. Nothing below interprets the results; the discussion is yours to write.

1. **`rl.group_var_es(pnl, books, 0.05)` will crash.** The levels argument must be a list or tuple, not a single number, so write `(0.05,)` with the trailing comma.
2. **No per-stock rows.** `group_var_es` only reports the books, the total, the sum of the books and the diversification benefit. The question also asks for each stock, so add `rl.var_es_table` on each `pnl` column.
3. **The margins don't match part (b).** `rl.fit_margins(ret)` only chooses between normal and t. If your part (a) choice included NIG for some stocks, the simulation has to use those same margins.
4. **The copula may not be the one you chose.** The code always fits a Gaussian copula. If you chose the t copula in part (b), you need the t-copula fit instead.
5. **`100_00` is 10,000.** That meets the "at least 5,000" requirement, but you probably meant `100_000`.

Also, `ret` has to be in the same stock order as `dollars`, and demeaned the same way as in parts (a)–(b). The code below reindexes it to make sure.

Corrected version (runs in about 2 seconds). Set `choice` and `COPULA` to your answers from parts (a) and (b):